

An Open-Source Pipeline for College Basketball Lineup Analytics

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Abstract

Although lineup data is an invaluable tool for college basketball programs across the country, lineup-level analytics remain difficult to access in college basketball, especially in the women's game, where public data is inconsistent and most advanced tools are locked behind paywalls. By harnessing publicly available data from the NCAA and ESPN, this project develops a solution to fill this gap: an open-source, fully automated pipeline that reconstructs true on-court lineups, computes transparent performance metrics, and visualizes them in an interactive R Shiny application. This system provides coaches with actionable insights that go far beyond traditional box-score statistics by integrating lineup efficiencies, individual player impact, and opponent scouting into a single platform. A case study on the 2024-25 UNC Men's Basketball team then illustrates the tool's practical value, revealing lineup strengths and matchup tendencies that could influence strategic decisions in both preparation and in-game situations. Overall, the pipeline offers accessible, reproducible solutions capable of serving both men's and women's programs, while also laying the foundation for future prescriptive features such as pre-game lineup optimization and real-time substitution recommendations.

1 Introduction

College basketball is a sport won on the margins. With six single-elimination games required to win a championship, plus a dominant regular season needed to secure a favorable path, teams must ensure they capitalize on every possible competitive advantage available to them. Coaches need to identify what actually makes their team successful, translate those insights into optimal player combinations, and use those lineups to shape favorable matchups and possession outcomes. Yet, across both men's and women's college basketball, the tools necessary to evaluate lineup effectiveness at a granular level remain limited, fragmented, or locked behind expensive paywalls. Easily accessible, traditional box-score statistics and team-wide efficiency metrics are too broad to guide nuanced rotation decisions, and they fail to answer the questions coaches need most in high-leverage moments: Which lineup should we trust? Against which opponent? And under what circumstances?

While existing lineup-based analytics platforms, such as EvanMiya and CBBanalytics on the men's side, or Hoop Explorer for women's basketball, provide pieces of the puzzle, they are often incomplete, opaque, or difficult to integrate into a real coaching workflow. Women's basketball, in particular, suffers from a substantial analytics gap with limited public data, inconsistent sources, and a nearly total absence of lineup-level tools. This lack of accessibility and transparency creates a real disadvantage for programs seeking an edge in preparation and in-game decision-making. These substantial limitations are what motivate our central research question: Can we develop a fully automated, open-source lineup analytics system that equips women's and men's collegiate basketball programs with transparent, actionable insights capable of improving lineup decisions, opponent scouting, and overall competitive performance? To answer this question, we built an end-to-end data pipeline that automatically scrapes NCAA and ESPN data, processes play-by-play events into clean and reliable lineup statistics, and visualizes these insights in an interactive, open-source R Shiny dashboard.

Although the tool was initially designed to fill the analytics void in women's basketball, its generalizable structure also makes it an immediately valuable resource for men's programs. A case study on the 2024-25 UNC Men's Basketball team presented later in this paper demon-

strates how this system reveals overlooked lineup efficiencies and opponent tendencies that could have meaningfully influenced tournament outcomes. By integrating lineup analysis and opponent scouting into a single, transparent platform, this project aims not only to close the analytics gap in women’s basketball but also to provide a powerful, accessible engine for data-driven decision-making across college basketball as a whole.

2 Background

Lineup and individual impact statistics are not a novel idea. However, these tools are far more developed on the men’s side than the women’s, and even the most prominent platforms suffer from major limitations. For example, many lineup datasets are locked behind paywalls, rely on opaque proprietary models, or provide only partial, surface-level functionality. Existing sites also lack transparency or offer limited reproducibility. Perhaps the most well-known of these sites is evanmiya.com. In addition to team ratings, individual player impact, and transfer portal analysis, Miya offers a “Lineup Metrics” tab. However, most of the details of this page are quite vague – “Advanced lineup performance visualizations”, “Lineup Ratings”, “Projections/Grades”, and “Lineup Analysis” are the available tabs and description of the service – and the statistics are behind a paywall. The lineup projections take an input of five players and predict offensive and defensive effectiveness going forward. This model relies on the individual player EvanMiya grades as well as the lineup’s observed performance throughout the current season (Miya, 2025).

CBBanalytics is another men’s college basketball website that provides some level of insightful features and tools. In terms of lineups and player impact, CBBanalytics provides four main features for an individual team, each presented in a separate tab on a team’s overview page. The first is “Lineups”, which allows a user to select which type of lineup they would like to view (1-player, 2-player,..., 5-player) and displays key statistics in tables, such as net rating and four factors. Additionally, it provides some charts to deepen a user’s understanding of a player’s minutes in a game. An example use case of these charts would be “Henri Veesar has been on the court for 100% of North Carolina’s shots during the 32nd minute of North Carolina’s games this season,” or “Henri Veesar has taken 14% of North Carolina’s shots during the 32nd

minute of North Carolina’s games this season.” The next tab is called “Combos”. This allows the user to choose a set of players and understand how the team performs with any subset of those players on the court. For example, through four games, North Carolina was +52.9 points per 100 possessions with Veesar, Caleb Wilson, and Seth Trimble on the floor in 40 minutes, but -31.5 points per 100 possessions with Trimble and Veesar on the floor and Wilson off the floor in 11 minutes. The “On-Off” tab allows a user to see which specific team metrics improve/regress when a player is on the court vs on the bench. For example, through four games, Kyan Evans is in the 98th percentile for the difference of the team’s AST/TO ratio when he is on the floor vs when he is off the floor. However, he is in the 7th percentile for the difference in the team’s DREB% when he is on vs when he is off the floor, allowing the user to see where specifically Evans’ skillset brings value to the team. The last tab, “Teammates”, enables users to analyze how a certain player’s presence on the floor affects the rest of the team’s individual statistics. For example, through four games, Veesar has played 86 minutes with Wilson on the floor and 29 minutes without him. With Wilson on the floor, Veesar has averaged 25.3 points per 40 minutes, compared to 13.7 with Wilson off the floor (Canova, 2025).

With this being said, these sites ultimately only provide insights for men’s basketball programs; there doesn’t seem to be many publicly available tools like CBBanalytics and EvanMiya for women’s college basketball. The website Hoop Explorer has lineup data with similar use cases to the CBBanalytics “Lineups” tab. Although the interface isn’t as user-friendly, a user can look through a team’s lineups, adjust for luck and garbage time, and filter for the quality of the opponent (Hoopexplorer, 2025). The wehoop R package allows users to easily access women’s college basketball play-by-play data (Saiem Gilani, 2025). However, it isn’t as clean as the data for men’s college basketball, and there are no helper functions that provide team strength metrics or player statistics, like the cbbdata package for men’s college basketball (Weatherman, 2025).

3 Methods

Due to the lack of an accessible and unified source for lineup-level data, especially for women’s basketball, our methodology focuses on building the full analytics pipeline from scratch. We

begin with automated scraping of NCAA and ESPN play-by-play logs, then reconstruct lineups and possessions with a set of cleaning and parsing procedures tailored to the inconsistencies in public data. These reconstructed events feed into our metric calculations, which are then visualized in an interactive R Shiny dashboard designed around real coaching workflows. An automation layer ties the system together, ensuring that lineup and opponent information stays current throughout the season. This end-to-end structure allows the tool to move seamlessly from raw web data to actionable insights.

3.1 Web-Scraping Pipeline

The first stage of this pipeline prioritizes the collection of the underlying play-by-play data. Because neither ESPN nor the NCAA provides all of the information required for lineup reconstruction, we use both sources in tandem. ESPN provides the official starters for each team, while the NCAA provides the full substitution strings needed to update the five players on the floor throughout the game. By combining the two data sources, we obtain the complete rotational information necessary to reconstruct every on-court lineup.

Our scraper harnesses Selenium to load dynamic ESPN and NCAA pages and BeautifulSoup to parse the HTML once rendered. For each game, the scraper first retrieves the starters from ESPN and stores them in a starters table. It then collects both ESPN’s and NCAA’s play-by-play logs, saving each as a raw CSV. NCAA’s PBP data is used as the primary event log, as it contains all substitution entries in recognizable string formats (e.g., “Luka Bogavac, substitution in; Caleb Wilson, substitution out”). These raw files form the foundation for the next step of the pipeline: translating textual substitutions and event markers into continuous lineup segments.

After collecting the raw logs, we reconstruct the full sequence of lineups by beginning with the ESPN starters vector and updating it according to each substitution listed in the NCAA’s PBP data. We filter the NCAA log to include only substitution rows, parse each substitution into its entering and exiting players, and update the on-court player vector accordingly. Each substitution includes a precise timestamp (period + clock), allowing the system to mark the start and end of each lineup interval. As the algorithm moves chronologically through the game, it produces a complete sequence of lineup segments with associated timestamps.

These reconstructed intervals are then combined with scoring and possession events to generate the core analytical file, which lists every unique lineup a team used in a game, the exact time it played, and the possessions associated with that interval. It is the key intermediate product that enables later stages of metric computation and visualization. Season-level aggregation functions then merge these per-game lineup_time files into comprehensive lineup datasets for all 1-player through 5-player combinations.

3.2 Metric Computation

Once lineup intervals and possessions are established, we compute the efficiency and performance metrics used throughout the dashboard. These include Offensive Rating (points scored per 100 possessions), Defensive Rating (points allowed per 100 possessions), Net Rating, Pace, Three-Point Efficiency, and the offensive and defensive Four Factors (EFG%, TOV%, ORB%, FTR).

Once these efficiency and performance metrics are established, the dashboard presents them in a series of intuitive tabs that guide coaches through progressively deeper layers of analysis. The interface moves from evaluating lineup strengths to understanding opponent tendencies, to isolating individual player impact, creating a unified system for data-driven game planning.

3.3 R Shiny App

Team Lineup Efficiency Tab

This tab allows the user to evaluate the efficiency of their team's lineups through a variety of visualizations and customizable filters. The lineup data is aggregated on a season-level and is available for all possible combinations of players (5-player combinations, 4-player combinations,..., 1-player combinations). The key visualizations include bar graphs for net rating, pace, and three-point efficiency, as well as a summary table that contains offensive four-factor data for the chosen subset of lineups. We decided to highlight net rating, pace, and three-point efficiency in their own bar graphs because each tells the user a lot about the quality of the team and their style of play, but they are very easy to customize and alter per a coach's request. In terms of filtering, subsets of lineups can be chosen by filtering for a set of players, or by selecting

a maximum or minimum of possessions and minutes. Lineup data can be quite noisy, so the time and possession filters allow coaches to quickly and easily find how their most-used lineups perform. In reality, a coach is not going to close a game without their team's star player, regardless of how well that player is playing. The player filtering feature lets a coach easily search for lineups that include their star player to determine which set of role players to put on the floor alongside them. Each visualization can be sorted by a metric, such as net rating. This allows a coach to have additional context within the visual. For example, when looking at the pace bar graph, a coach might also want to consider the net rating of the lineup, as pace doesn't help to describe a lineup's quality, just its playstyle. The coach can then sort the graph by net rating. Additionally, the bar graphs are interactive, and hovering over each bar provides additional statistics for each lineup.

Overall Opponent Data Tab

To make a coach's life easier, the overall opponent data tab helps with opponent scouting, keeping it in the same place as the lineup data, rather than in another tool. This data is pulled from the `cbbdata` package by Andrew Weatherman, which is available only for men's college basketball. Currently, this tab is only functional for men's due to the lack of a similar data source for women's college basketball.

A user can select a team and a year to investigate, and the page returns three tables. The first of which is a team strength table. With statistics such as BPI, projected seed, quality wins, and strength of schedule, a coach can identify the quality of opponents they are facing and what level of game plan they will have to construct if they expect to beat their opponent. However, this table is only available for the current season, as this is the way the `cbbdata` function we are using to pull the data is constructed.

The next table displays a team's performance over the course of the season, segmented by month. It contains ten metrics: opponent-adjusted offensive efficiency, opponent-adjusted defensive efficiency, offensive four factors, and defensive four factors. When preparing for a late-season game, such as an NCAA tournament game, a coach might be wise to pay more attention to how a team has played in January and February, rather than November, since a team might have had a hot or cold shooting stretch early in the season, played a soft out-of-conference

schedule, or changed the makeup of their rotation. Perhaps a team's star player suffered an injury in December and missed the entire month of January. It makes sense to evaluate an opponent based on how they performed when their star player was in the lineup, and ignore the month of January.

The final table contains individual player statistics, including traditional box score numbers, shooting efficiency and tendencies, and impact metrics. This table allows a coach to get a quick glance at a team's roster before diving into their film study. Projected players to gameplan against are bolded. These include:

- High usage stars: BPM ≥ 5.5 , USG% ≥ 23.0 , mins/game ≥ 25
- Efficient stars/stars in their role: BPM ≥ 6.5 , TS% ≥ 65 , mins/game ≥ 15
- Other high-impact players: PORPAG ≥ 4.0 (points over replacement player per (adjusted) game)

The bolding helps direct a coach's focus to these players, allowing them to start their game planning with those players. The shooting tendencies, combined with efficiencies, can enable a coach to identify which shots the team should try to prevent a certain player from taking, and which shots might be beneficial if the player takes more of them.

Individual Player Impact Tab

When performing a holistic team evaluation, a coach might identify an overarching weakness for the team as a whole. For example, the team might consistently turn the ball over too much. It's useful for a coach to have a tool that allows them to determine whether a player's team is turning the ball over more or less when they're on the court than when they're off. By performing some data manipulation on the 1-player lineups to filter for lineups when that player is on the court versus lineups without that player, we can answer that question.

A user can select a player from any available team's season and view a team's net rating, offensive and defensive ratings, offensive four factors, true shooting percentage, and assist percentage when that player is on the floor, off the floor, and the differences. Coaches can assess their players' skill set by seeing which specific metrics they help with on the floor and which they hurt.

3.4 Automation

The final stage of our methodology is an automation layer designed to keep the system updated throughout the season without requiring manual intervention. The functions `update_season_summary_table()` and `update_new_games_and_aggregate()` serve as the core update engine. The first checks for newly played games and regenerates season-level metrics, while the second scrapes and processes only the games not yet stored in the database, avoiding redundant computation and dramatically reducing runtime. These functions are integrated into a GitHub Actions workflow that runs the update pipeline on a scheduled basis (every 24 hours). The workflow automatically pulls the most recent code and data, executes the update functions, regenerates the necessary CSVs and season summaries, and commits the updated dataset back to the repository. This continuous integration structure ensures that lineup and opponent data remain current without the user having to manually run any scripts, eliminating the risk for user error.

Throughout this multi-stage process, our methodology transforms inconsistent public play-by-play logs into reliable, fully open-source analytics capable of supporting lineup decisions, player evaluation, and opponent scouting.

4 Results

The 2024-25 season was a major disappointment for the University of North Carolina's men's basketball team. An 11-seeded first-round exit was not what many Tar Heel fans expected, with the roster consisting of reigning ACC Player of the Year RJ Davis, two five-star freshmen in Ian Jackson and Drake Powell, and a returning supporting cast from the previous year's 29-8, 1-seeded team, which included Seth Trimble, Elliot Cadeau, and Jae'lyn Withers. To demonstrate our app's functionality, we will walk through some realistic use cases from the perspective of the 2024-25 North Carolina Tar Heels men's basketball team.

In 2023-24, UNC started the lineup of RJ Davis, Elliot Cadeau, Cormac Ryan, Harrison Ingram, and Armando Bacot 29 times. In 2024-25, UNC's most-used starting lineup started just 10 games together. Throughout the season, head coach Hubert Davis continuously tinkered with the starting lineup. The starting lineup coach Davis rolled out at the end of the season

was RJ Davis, Elliot Cadeau, Drake Powell, Jae'Lyn Withers, and Ven-Allen Lubin. How did this lineup perform? We can use the Lineup Efficiency Table on the Team Lineup Efficiency Tab to answer this question. Sorting by possessions, we can see that this starting lineup played more possessions than any other lineup Davis used all season, and boasted an impressive +31.2 net rating. We can also analyze where the high-octane offensive performance came from. The lineup was not particularly elite at generating offensive rebounds or preventing turnovers and was poor at generating free throws, but knocked down shots at an extremely high rate.

Key Stat Visualization		Lineup Efficiency Table							
Lineup	Minutes	↓ Possessions	Net Rating	ORTG	DRTG	Offensive Four Factors			
						eFG%	TOV%	FTR	OREB%
('Drake Powell', 'Elliot Cadeau', 'Jae'Lyn Withers', 'RJ Davis', 'Ven-Allen Lubin')	110.3	179.5	31.2	125.3	94.1	61.4	16.2	20.2	24.1
('Drake Powell', 'Elliot Cadeau', 'Jae'Lyn Withers', 'RJ Davis', 'Ven-Allen Lubin')	93	165.2	28.5	119.9	91.4	52	10.9	43.2	3

Figure 1: Statistics for UNC's most used lineup in 2024-25

Switching over to the bar graph for net rating, we can see whether Davis' most-played lineup was his best.

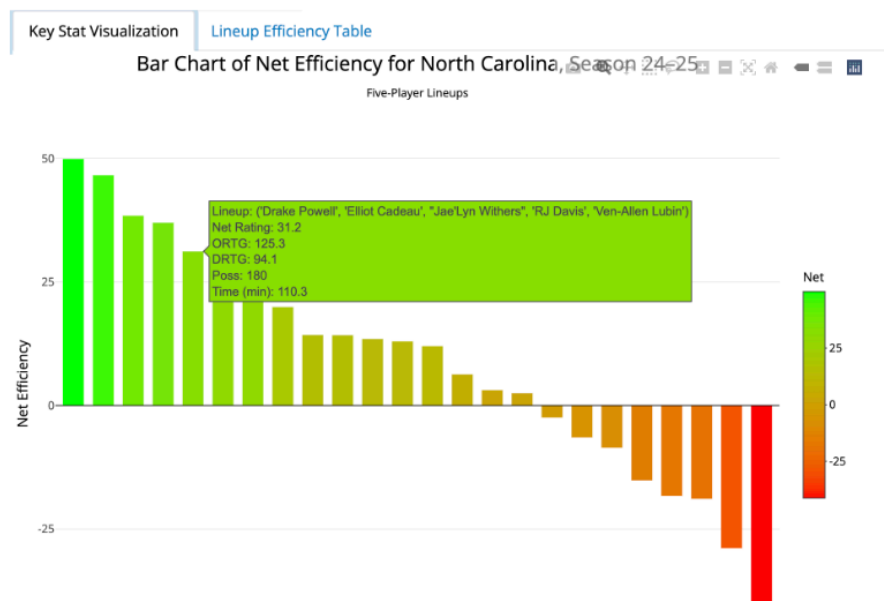


Figure 2: UNC's 2024-25 5-man lineups, sorted by net rating

A few lineups outperformed this lineup. It turns out that inserting Seth Trimble instead of Elliot Cadeau into that lineup was UNC's most efficient lineup throughout the season, albeit over a sample size of less than half of the RJ Davis, Elliot Cadeau, Drake Powell, Jae'Lyn Withers, and Ven-Allen Lubin lineup. At the time of writing this paper, Elliot Cadeau has played four games with Michigan in the 2025-26 season. His BPM is currently just 0.6, his eFG% is 44.6%, and his TOV% is 30.1%, all poor numbers from a former five-star recruit starting on a national championship contender. Hubert Davis might have benefited from playing Cadeau less than he did last season.

UNC found itself in many huge deficits last season. Because of this, they often had to push the pace and shoot more threes to get back into the game. It turns out that the lineup that played the fastest (minimum 30 possessions) was also the lineup that shot the highest percentage from three: RJ Davis, Elliot Cadeau, Seth Trimble, Drake Powell, and Jae'Lyn Withers.

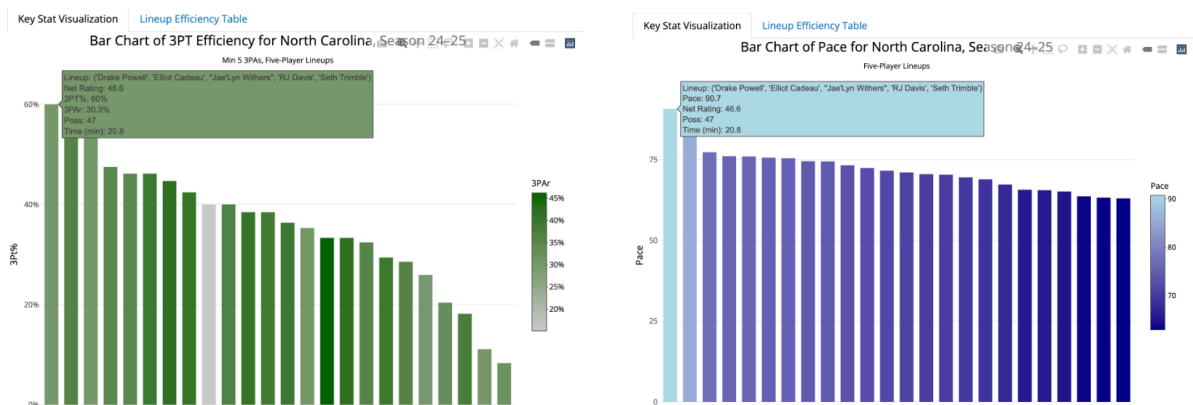


Figure 3: UNC's fastest-paced lineup was also their most prolific three-point shooting lineup, albeit in a small sample size

This lineup also had an impressive +46.6 net rating. Why didn't Davis lean on this lineup more? Given the lineup consisted of three guards, two wings, and no bigs, one might hypothesize that this lineup struggled on the glass.

Key Stat Visualization		Lineup Efficiency Table								
Possessions	Net Rating	ORTG	DRTG	Offensive Four Factors				3PAr	↓ Pace	
				eFG%	TOV%	FTR	OREB%			
47.2	46.6	120.8	74.2	51.5	8.5	90.9	16.7	30.3	90.7	
40.8	-2.4	80.8	83.3	32.9	17.1	29	36	31.6	85.7	

Figure 4: Statistics for the Davis/Cadeau/Trimble/Powell/Withers lineup

A glance at the Lineup Efficiency Table supports this hypothesis. While this lineup was effective in short spurts, playing it for extended stretches wasn't realistic due to the poor rebounding. Additionally, it's extremely unlikely that this lineup would sustain such a low turnover rate, and the meager 51.5% eFG% suggests that the lineup struggled to connect on two-point shots, since we'd expect a lineup with a 60% 3PT% to have a much better eFG%.

After dropping back-to-back games by one point to Stanford and Wake Forest in mid-January, UNC sat at 12-8 and needed to close the season strong to make the tournament. Hubert Davis made a change to the starting lineup, inserting Ven-Allen Lubin. UNC went 10-5 in the rest of the regular season and the ACC tournament, with three of those losses coming to Duke. Lubin had a massive impact on the team, as shown on the Individual Player Impact Tab.

Ven-Allen Lubin On/Off Splits

SPLIT	POSSESSIONS	MINUTES	ORTG	DRTG	NET RATING	EFG%	TS%	AST%	TOV%	OREB%	FTR
On	1,230.2	694.7	117.5	101.1	16.3	56.2	60.1	22.2	16.0	26.1	33.9
Off	1,327.6	765.3	112.4	110.2	2.2	52.7	56.7	20.1	14.8	26.4	36.7
Diff	-	-	+5.1	-9.1	+14.2	+3.5	+3.4	+2.1	+1.2	-0.2	-2.7

Figure 5: Ven-Allen Lubin's 2024-25 On/Off Splits

UNC was 14.2 points per 100 possessions better when Lubin was on the floor compared to when he was off the floor. Lubin's efficiency stood out, with a 68.4% eFG% and a 69.4% TS%, both by far the highest of any UNC player who played real minutes. This translated into improved overall team efficiency, as UNC's eFG% and TS% jumped noticeably with Lubin on the floor.

After UNC's victory over San Diego State in the first four, the Tar Heels faced off against Mississippi in the Round of 64. UNC was favored to win this game, and many Tar Heel fans envisioned a Cinderella run to the Sweet 16. However, UNC fell into an 18-point halftime deficit, and their furious comeback was thwarted, a common theme of the season. What went wrong for the Tar Heels, and was it preventable? We can investigate these questions using the Overall Opponent Data Tab.

2025 Mississippi Team Strength Stats (ESPN)											
BPI	RANK	OFF BPI	RANK	DEF BPI	RANK	SOR RANK	SOS RANK	QUALITY WINS	QUALITY LOSSES	PROJECTED WINS	PROJECTED SEED
14.1	29	7.4	26	6.7	33	16	20	12	11	24	4

Figure 6: Mississippi's 2024-25 resume (ESPN)

For starters, we can see that while UNC was the odds-on favorite to win this game, Mississippi wasn't exactly a weak 6-seed. ESPN projected Mississippi to receive a 4-seed, and it was one of 10 teams to finish in the top 25 in KenPom offensive and defensive efficiency. Mississippi was also a different team at the end of the season than it was at the start of the year.

MONTH	ADJUSTED OE	ADJUSTED DE	Offensive Four Factors				Defensive Four Factors			
			EFG%	TO%	OREB%	FTR	EFG%	TO%	OREB%	FTR
Nov	118.1	97.9	52.3	11.6	39.9	37.5	51.3	23.8	26.9	36.7
Dec	113.5	93.8	53.7	14.3	26.9	33.5	45.0	23.1	30.4	37.3
Jan	110.3	88.8	50.0	13.8	21.8	32.0	45.2	21.3	31.9	37.7
Feb	122.7	103.9	51.0	11.2	20.5	43.3	55.7	14.7	32.3	48.4
Mar	125.9	97.3	50.7	13.4	27.4	34.3	54.5	18.4	29.0	38.6

Figure 7: Mississippi's offensive efficiency improved throughout the season

In February and March, Mississippi's offense improved significantly compared to December and January. Much of the February offensive success can be attributed to the increase in Mississippi's free-throw rate, and that carried over to the game. Mississippi had a 42.6% FTR compared to UNC's 25.4%. As we can see from the four-factor data, Mississippi was one of the worst rebounding teams in the tournament, and yet they crushed the Tar Heels on the glass in this game. In fact, Mississippi ranked outside the top 300 in offensive OREB% and top 200 in defensive OREB% in the 2024-25 season. However, we don't think these problems are necessarily because of UNC's game plan. UNC forward Jae'Lyn Withers played just 16 minutes before exiting the game with an injury.

Jae'Lyn Withers On/Off Splits

SPLIT	POSSESSIONS	MINUTES	ORTG	DRTG	NET RATING	EFG%	TS%	AST%	TOV%	OREB%	FTR
On	1,121.9	629.6	118.5	105.0	13.5	55.8	60.9	21.7	15.5	24.8	40.8
Off	1,435.9	830.4	112.0	106.5	5.5	53.2	56.4	20.6	15.3	27.3	31.4
Diff	—	—	+6.5	-1.5	+8.0	+2.6	+4.5	+1.1	+0.2	-2.5	+9.4

Figure 8: Jae'Lyn Withers' 2024-25 On/Off Splits

The Tar Heels were far more efficient offensively with Withers on the floor, particularly because of his ability to get to the free-throw line. While Withers didn't produce on the offensive glass, he led UNC in DREB%, and his impact on the defensive glass was greatly missed.

2025 Mississippi Player Statistics

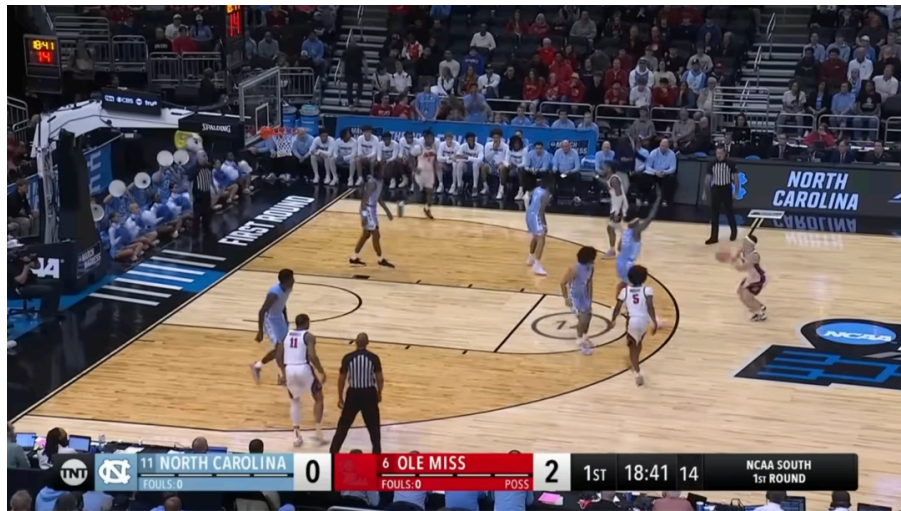
Min 25 minutes played

PLAYER	ROLE	EXP	Basic Stats										Shooting Accuracy/Tendencies					
			MPG	PPG	RPG	APG	TOV	SPG	BPG	USAGE RATE	EFG	TS	RIM FG%	RIM ATTEMPT RATE	MIDRANGE FG%	MIDRANGE ATTEMPT RATE	3PT%	3PT ATTEMPT RATE
Sean Pedulla	Scoring PG	Sr	32.8	15.4	3.5	3.8	2.0	1.9	0.2	23.6	54.4	58.0	57.5	26.7	39.3	21.0	39.2	52.4
Jaylen Murray	Scoring PG	Sr	30.5	10.3	2.2	3.4	1.3	1.4	0.2	18.3	50.0	53.2	56.1	17.9	37.4	31.1	37.0	50.5
Matthew Murrell	Combo G	Sr	30.3	10.8	3.5	1.8	0.6	1.8	0.2	17.1	50.2	54.3	60.5	27.3	43.3	24.1	31.9	48.6
Dre Davis	Stretch 4	Sr	26.0	10.1	4.8	1.3	1.2	1.0	1.0	21.1	50.0	53.2	58.2	35.6	43.0	36.4	32.5	28.6
Jaemyn Brakefield	Wing F	Sr	23.3	11.1	4.3	1.8	1.2	0.8	0.2	23.0	56.1	60.4	68.1	45.4	39.1	26.3	35.1	28.2
Malik Dia	Stretch 4	Jr	20.9	10.8	5.7	0.8	0.8	0.7	0.9	25.2	51.5	54.0	66.1	41.8	31.2	30.6	34.5	27.6
Davon Barnes	Wing G	Sr	20.5	5.0	3.2	0.8	0.5	0.7	0.3	14.2	42.6	47.6	66.7	29.6	33.3	29.6	21.2	40.7
Mikeal Brown-Jones	PF/C	Sr	10.1	4.1	1.6	0.4	0.6	0.3	0.5	21.3	49.3	57.8	60.9	64.8	28.6	19.7	18.2	15.5
Eduardo Klafke	Wing G	Fr	10.0	1.4	1.1	0.4	0.3	0.4	0.2	7.9	62.5	61.9	60.0	13.9	0.0	11.1	48.1	75.6
Robert Cowherd	Scoring PG	So	4.2	0.0	0.2	0.2	0.1	0.2	0.1	6.5	0.0	0.0	0.0	25.0	NA	0.0	0.0	75.6

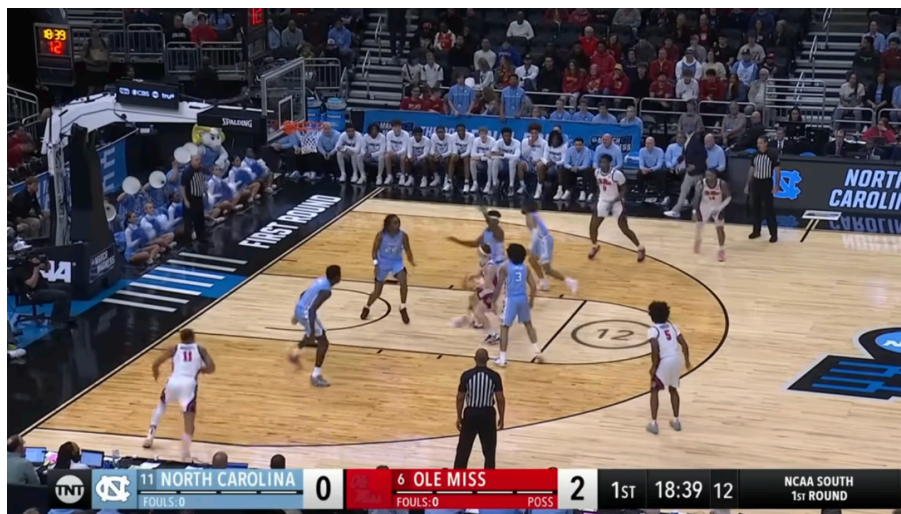
Figure 9: Sean Pedulla was Mississippi's most dangerous player in 2024-25

From our player table on the Overall Opponent Data Tab, we can see which players were highlighted as targets for game planning. It highlighted Sean Pedulla, a dynamite shooter and underrated playmaker. Pedulla is a high-volume shooter with elite efficiency, shown by his 52% 3PAr and 39.2% 3PT%. He was more efficient outside the arc than inside, suggesting an obvious strategy: run him off the three-point line. In addition, Pedulla is more than capable of making skip-passes to shooters in the corner and finding his bigs rolling to the basket. Another emphasis would be to not overhelp and let Pedulla challenge the Tar Heel bigs at the rim. He's adept at drawing fouls, but forcing him into mid-range shots, runners, and contested layups is beneficial for the defense. Pedulla had 20 points, which led all scorers, with 5 assists to go with

5 turnovers. The first part of the game plan was rather sound; Pedulla had a 3PA% of just 28.6%. Here, Lubin closes out aggressively on Pedulla, preventing him from shooting a three.



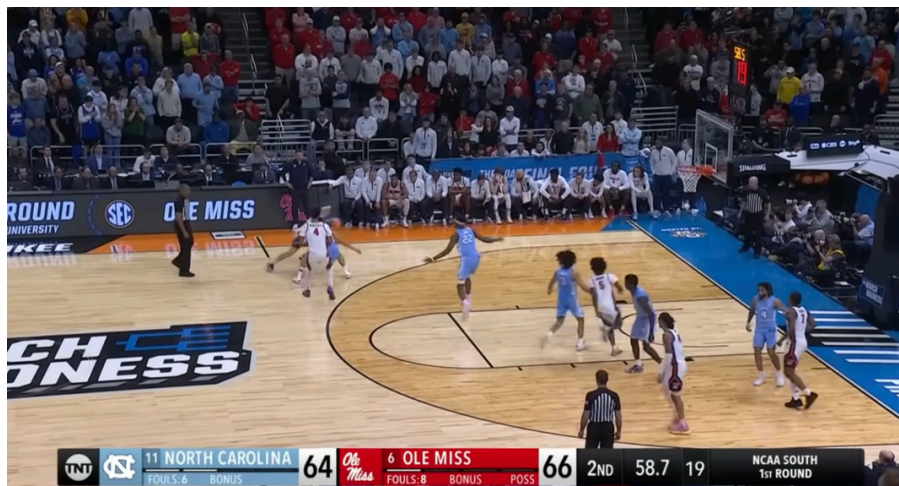
However, when Pedulla enters the paint, the entire defense collapses. RJ Davis overhelps and gets lost when Dre Davis relocates. Dre Davis is classified as a Stretch 4 in the player table. While his percentage isn't elite, it's poor defensive discipline from RJ Davis.



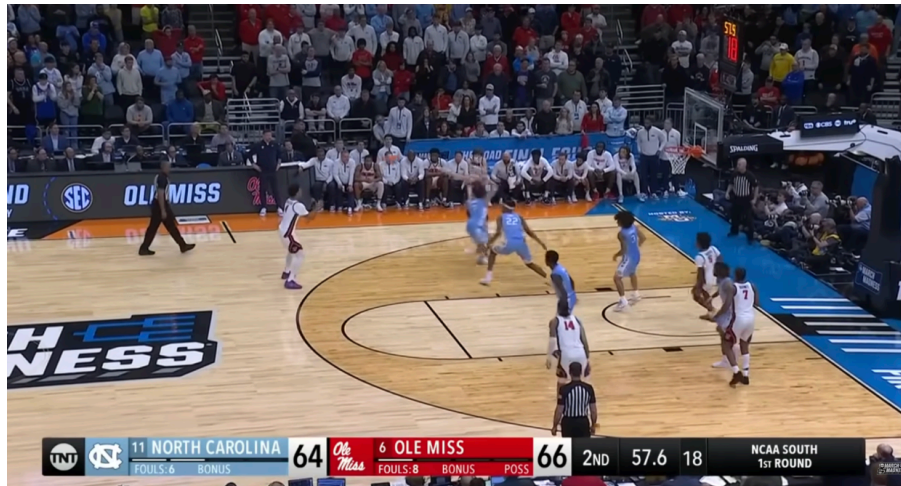
Cadeau can't close to Dre Davis fast enough, and Malik Dia screens RJ Davis, leaving Dre Davis wide open.



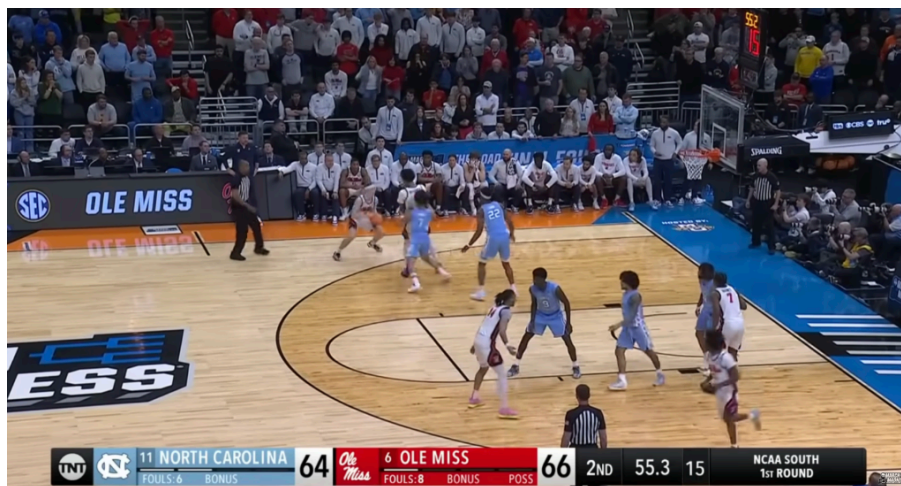
On the defining defensive possession of the game, Pedulla is being guarded by Seth Trimble, UNC's best perimeter defender. He rejects a ball screen from Jaemyn Brakefield.



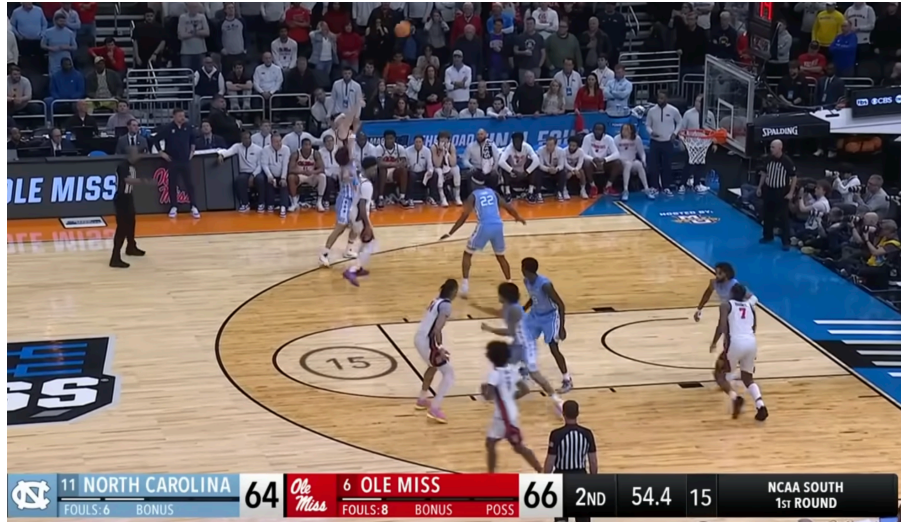
Lubin ignores Brakefield and helps on Pedulla. This is a defensible decision. While Brakefield is a capable shooter, Lubin knows that Mississippi does not want him shooting an above-the-break three with 18 seconds on the shot clock in the final minute of a tournament game. It seems like UNC is employing a strategy of “anyone but Pedulla”.



Lubin is correct, and Brakefield doesn't even consider shooting the three, and instead flows into a dribble handoff to get the ball back to Pedulla. However, Lubin is far too low, not even attempting to help Trimble.



Brakefield doesn't even roll – the action was designed for Pedulla to take the shot, and Lubin fails to help Trimble run him off the line. Trimble can't fight over the screen, and Pedulla has enough space to hit the dagger, his second three of the game.



From those two plays, we can see some of the problems that cost UNC’s chances of salvaging a disappointing season, and we can see how UNC differed from our recommendations based on our opponent scouting analysis, whether that was due to a differing game plan or a lack of execution.

5 Conclusion

This project set out to address the absence of transparent, accessible lineup-level tools, particularly on the women’s side in college basketball. By building an end-to-end pipeline that automates data collection, reconstructs true on-court lineups, computes interpretable performance metrics, and presents them in a unified, coach-oriented dashboard, we demonstrate that open-source tools can meet a level of analytical depth that has previously been restricted to paywalled platforms. The system not only fills a structural gap in women’s basketball but also proves its practical value through the presented case study on the 2024-25 UNC Men’s Basketball season, where it revealed lineup tendencies and matchup insights that wouldn’t have been as obvious using traditional statistics alone. The results demonstrate the importance of integrating lineup analytics, individual player impact, and opponent scouting into a unified framework. Coaches can quickly identify which combinations generate the highest net ratings, which players can meaningfully alter team efficiency, and which opponent trends require special preparation. Overall, the case study highlights how even a high-profile program with ample resources could benefit from the additional clarity and specificity produced by this system.

Although the pipeline already manages the technical challenges of scraping inconsistent public data and reconstructing lineups with a high degree of reliability, one important limitation remains: coaches must still interpret the insights themselves. The dashboard shows what the best lineups are, when players help or hurt particular metrics, and how opponents' tendencies evolve across the season, but converting those insights into optimal pre-game plans or mid-game adjustments still requires a manual, human-driven process. This limitation naturally points toward the main future direction for the project: the development of a lineup suggestion engine. A pre-game version of this engine could combine opponent efficiencies, personnel tendencies, and matchup data to recommend optimal lineups before tip-off. Furthermore, a mid-game engine, based on live play-by-play data, could track possession-by-possession dynamics and recommend substitutions or lineup changes in real time. The human-guided analysis demonstrated in our case study suggests that such a system would be both feasible and impactful, and as a result, formalizing it algorithmically would represent a significant step forward in accessible basketball analytics.

Ultimately, this project shows how an open-source and reproducible pipeline can transform fragmented public data into actionable lineup insights for both men's and women's basketball. By combining transparent methodology with practical coaching applications, the system has laid the foundations for future tools that can actively guide decision-making in real time. As data infrastructures improve and engine technology becomes more robust, the potential for open-source analytics to meaningfully influence college basketball strategy will only continue to grow.

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